

WASP-52b

R-0338

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_221025 · created 2026-07-18T06:23:33+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459877.6829 +/- 0.0072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.215 +/- 0.062
Transit depth (Rp/Rs)^2	4.62 +/- 2.66 [%]
Orbital Inclination (inc)	87.45 +/- 2.82
Airmass coefficient 1 (a1)	0.18 +/- 0.097
Airmass coefficient 2 (a2)	1.5 +/- 1.28
Scatter in the residuals of the lightcurve fit is	55.79 %
Best Comparison Star	None
Optimal Aperture	2.79
Optimal Annulus	5.8
Transit Duration (day)	0.083 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.215 ± 0.062 vs 0.1646 (deviation 0.8σ)
Duration fit vs published	0.083 d vs 0.0758 d (deviation 0.44σ)
Mid-time O–C	17.5 min
Depth SNR	3.4
Residual scatter	55.79 %

Light curve

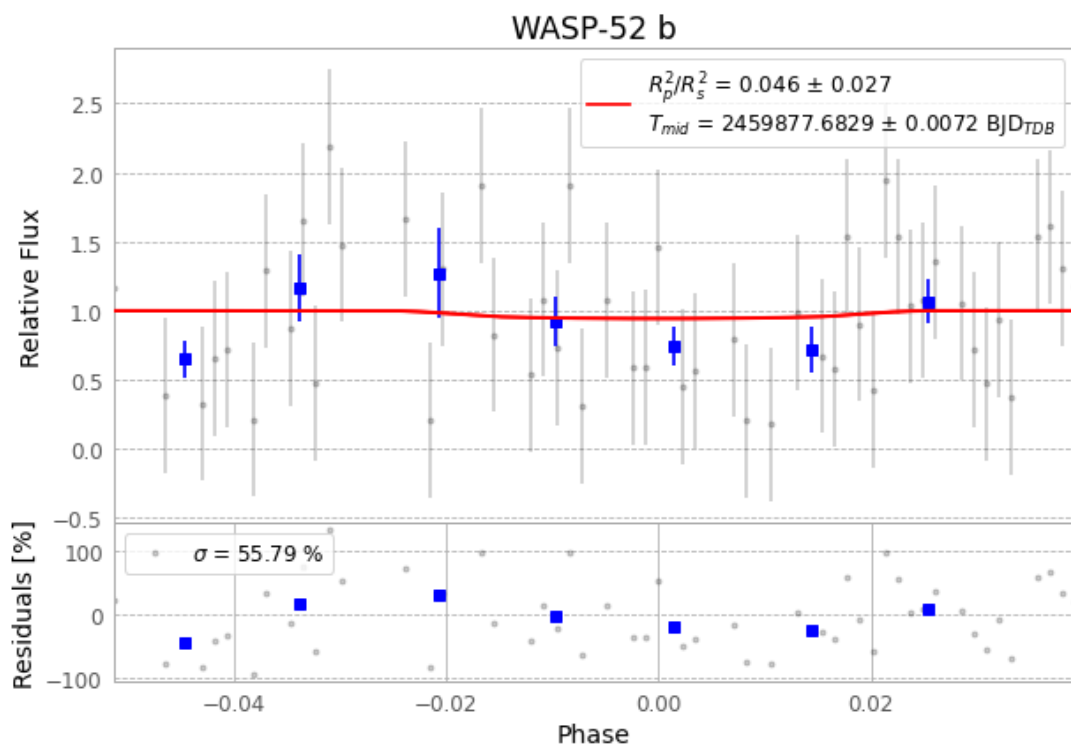


Figure 1 — FinalLightCurve_WASP-52 b_2022-10-24.png (SHA-256 0de6b8de43404279...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [231, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 51542cb04a9efcb8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

76 FITS frames (50 MB), WASP-52221025020613.FITS → WASP-52221025055415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-221025.log (4391ac27cc4c...), FinalLightCurve_WASP-52 b_2022-10-24.png (0de6b8de4340...), FinalLightCurve_WASP-52 b_2022-10-24.csv (40e50542ce60...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 06:23Z.